EXPRIMENT 6

Question:

Write the python program for Vacuum Cleaner Problem

Aim:

To write a Python program for the vacuum cleaner problem, where an agent cleans two rooms (A and B) by sensing its location and whether the room is clean or dirty.

Algorithm:

1. Start.
2. Take the status of Room A and Room B (Clean or Dirty) as input.
3. Take the starting location of the vacuum cleaner (A or B).
4. Check the current room:

    I. If the room is **Dirty**, then **Suck** (clean it).
    II. Else if the location is **A**, then **Move Right** to B.
    III. Else if the location is **B**, then **Move Left** to A.

5. Update the room status or location after the action.
6. Repeat steps 4 and 5 until both rooms are Clean.
7. Stop.

Code:

lab6.py - C:/Users/ELCOT/AppData/Local/Programs/Python/Python314/lab6.py (3.14.6)

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```
# Vacuum Cleaner Problem - Simple Reflex Agent (2 rooms: A and B)

def vacuum_agent(location, status):
    if status == "Dirty":
        return "Suck"
    elif location == "A":
        return "Move Right"
    else:
        return "Move Left"


rooms = {}
rooms["A"] = input("Enter status of Room A (Clean/Dirty): ").capitalize()
rooms["B"] = input("Enter status of Room B (Clean/Dirty): ").capitalize()
location = input("Enter starting location (A/B): ").upper()

print("\nInitial State:", rooms, "| Location:", location, "\n")

step = 1
while rooms["A"] == "Dirty" or rooms["B"] == "Dirty":
    status = rooms[location]
    action = vacuum_agent(location, status)
    print("Step", step, ": Location =", location, ", Status =", status, ", Action =", action)

    if action == "Suck":
        rooms[location] = "Clean"
    elif action == "Move Right":
        location = "B"
    else:
        location = "A"

    step += 1

print("\nFinal State:", rooms)
print("Goal reached: Both rooms are clean!")
```

Output:

IDLE Shell 3.14.6

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```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
==== RESTART: C:/Users/ELCOT/AppData/Local/Programs/Python/Python314/lab6.py ===
Enter status of Room A (Clean/Dirty): Dirty
Enter status of Room B (Clean/Dirty): Dirty
Enter starting location (A/B): A

Initial State: {'A': 'Dirty', 'B': 'Dirty'} | Location: A

Step 1 : Location = A , Status = Dirty , Action = Suck
Step 2 : Location = A , Status = Clean , Action = Move Right
Step 3 : Location = B , Status = Dirty , Action = Suck

Final State: {'A': 'Clean', 'B': 'Clean'}
Goal reached: Both rooms are clean!
>>>
```

Result :

The program ran successfully.

- The vacuum cleaner started in Room A. Both rooms were dirty.
- It cleaned Room A first.
- Then it moved to Room B.
- It cleaned Room B also.
- Now both rooms were clean, so the program stopped.